

Binary	Bug ID	Ember-IO-Fuzzing			Fuzzware			Fuzzware-Icicle			SplITS			Hoedur			MultiFuzz		
		Med. R	Med. T	Hit	Med. R	Med. T	Hit	Med. R	Med. T	Hit	Med. R	Med. T	Hit	Med. R	Med. T	Hit	Med. R	Med. T	Hit
3DprinterX	FP_FW39	03:30	03:30	60%	00:09	00:09	100%	00:22	00:22	100%	00:07	00:07	100%	00:05	00:05	100%	00:01	00:01	90%
CNCX	FP_E02	00:21	—	10%	00:40	—	0%	00:51	—	10%	00:22	—	0%	00:08	—	30%	00:03	—	10%
	FW11	—	—	10%	—	—	50%	05:39	05:39	60%	12:20	12:28	70%	00:08	00:09	100%	01:34	04:48	90%
	FW18	00:20	17:25	60%	00:31	11:23	80%	00:33	07:50	90%	00:17	16:25	80%	00:07	00:17	100%	00:03	01:47	100%
ConsoleX	I01	—	—	0%	—	—	0%	—	—	0%	00:01	00:01	100%	00:07	00:07	100%	00:09	00:09	100%
GPSTrackerX	FW30	00:10	00:20	90%	00:07	00:07	100%	00:07	00:09	100%	00:06	00:12	100%	00:05	00:06	100%	00:01	05:17	70%
	FW31	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
	MF02	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
	MF03	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
	S01	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
	S02	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
GatewayX	E01	00:14	00:27	100%	02:03	—	30%	—	—	20%	01:43	—	10%	—	—	0%	05:15	—	10%
	FP_FRB01	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	50%
	FP_FRB02	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	50%
	FP_FRB03	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	14:51	—	0%
	FP_FRB04	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
	FP_FRB10	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	10%
	FP_FW21	00:00	00:00	100%	00:01	00:01	100%	00:01	00:01	100%	00:01	00:02	100%	00:06	00:07	80%	00:00	00:00	100%
	FP_FW22	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	14:51	14:51	60%
	FW12	00:11	03:10	100%	00:18	—	30%	00:17	—	50%	00:57	—	50%	—	—	0%	03:52	09:03	70%
	FW23	06:31	—	10%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	13:14	—	0%
PLCX	MF01	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	12:11	—	10%
	FP_FW25	00:00	00:00	100%	00:00	00:00	100%	00:01	00:01	100%	00:02	00:02	100%	00:03	00:03	80%	00:00	00:00	100%
	FW14	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	40%	—	—	0%
	FW15	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	30%	—	—	10%
	FW16	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
app_controlX	FW17	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	20%	—	—	0%
	FP_FRB53	—	—	0%	—	—	N/A	00:01	00:01	100%	—	—	N/A	00:05	00:05	100%	00:01	00:01	100%
	FP_FRB54	00:00	00:00	100%	—	—	N/A	00:01	00:01	100%	—	—	N/A	00:03	00:03	100%	00:00	00:00	100%
	FP_FRB55	—	—	0%	—	—	N/A	00:01	00:01	100%	—	—	N/A	00:05	00:05	100%	00:01	00:01	100%
	FRB48	—	—	0%	—	—	N/A	—	—	0%	—	—	N/A	—	—	0%	—	—	0%
	FRB49	—	—	0%	—	—	N/A	—	—	0%	—	—	N/A	—	—	0%	—	—	0%
	FRB50	—	—	0%	—	—	N/A	—	—	0%	—	—	N/A	—	—	0%	—	—	0%
betaflightX	FRB51	—	—	0%	—	—	N/A	—	—	0%	—	—	N/A	—	—	0%	—	—	0%
	FRB52	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
	FRB54	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
	FRB55	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
	FRB56	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%

Table 1: Fuzzing Results (Part 1 of 3)

Binary	Bug ID	Ember-IO-Fuzzing			Fuzzware			Fuzzware-Icicle			SplITS			Hoedur			MultiFuzz			
		Med. R	Med. T	Hit	Med. R	Med. T	Hit	Med. R	Med. T	Hit	Med. R	Med. T	Hit	Med. R	Med. T	Hit	Med. R	Med. T	Hit	
contiki-hello-4-4X	FW58	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
	H03	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
	H04	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
	H06	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
	H07	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
	H08	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
	H09	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
	H10	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
contiki-hello-4-8X	FRB20	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	00:58	—	—	0%
	FW58	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%		—	—	0%
	H03	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%		—	—	0%
	H04	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%		—	—	0%
	H06	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%		—	—	0%
	H07	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%		—	—	0%
	H08	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%		—	—	0%
	H09	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%		—	—	0%
	H10	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%		—	—	0%
	H11	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%		—	—	0%
	H12	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
contiki6lowpanX	FRB14	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
	H02	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
contiki_shellX	S06	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	01:14	01:24	100%	
hoverX	FRB38	—	—	20%	—	—	0%	—	—	0%	—	—	0%	01:03	01:03	100%	05:56	05:56	100%	
	FRB39	—	—	20%	—	—	0%	—	—	0%	—	—	0%	01:16	01:16	100%	07:28	08:22	100%	
	FRB46	—	—	20%	—	—	0%	—	—	0%	—	—	0%	06:55	06:55	80%	09:23	09:23	90%	
ip64-routerX	H13	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
loramacX	FP_FRB25	—	—	0%	—	—	0%	—	—	0%	—	—	0%	02:08	02:08	100%	—	—	0%	
	FP_FRB26	—	—	0%	—	—	0%	—	—	0%	—	—	0%	02:59	02:59	70%	—	—	0%	
	FP_FRB40	00:00	00:00	100%	00:12	00:12	100%	00:11	00:11	100%	00:09	00:09	100%	00:14	00:14	100%	00:01	00:01	100%	
	FP_FRB58	—	—	0%	—	—	0%	—	—	0%	—	—	0%	02:30	02:30	80%	—	—	0%	
	H18	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	10%	—	—	0%	
snmp-serverX	H14	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
	H15	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
	H16	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
	H17	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
zephyr-3330X	H28	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
zephyr-btX	FP_FRB41	—	—	0%	02:34	—	30%	03:34	—	0%	04:54	—	30%	00:46	00:48	100%	00:11	04:07	70%	
	FW47	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	
	FW48	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	

Table 2: Fuzzing Results (Part 2 of 3)

Binary	Bug ID	Ember-IO-Fuzzing			Fuzzware			Fuzzware-Icicle			SplITS			Hoedur			MultiFuzz		
		Med. R	Med. T	Hit	Med. R	Med. T	Hit	Med. R	Med. T	Hit	Med. R	Med. T	Hit	Med. R	Med. T	Hit	Med. R	Med. T	Hit
zephyr-f49ziX	FP_FRB44	—	—	0%	00:38	06:05	70%	—	—	0%	00:37	02:31	100%	01:45	01:45	100%	00:01	00:07	100%
	FP_FRB45	00:19	—	0%	00:20	—	10%	00:18	—	0%	00:16	—	20%	01:48	10:10	70%	00:01	—	30%
	H31	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
zephyr-nrf	FW54	—	—	0%	—	—	0%	—	—	0%	—	—	0%	16:28	16:35	80%	—	—	10%
	H33	00:00	—	0%	00:01	—	0%	—	—	0%	00:01	—	0%	00:09	—	0%	00:01	—	0%
	H34	00:00	—	0%	00:01	00:50	100%	—	—	0%	00:02	01:04	100%	00:11	05:27	70%	00:01	—	20%
	H35	00:00	—	0%	00:01	—	0%	00:01	—	0%	00:01	—	0%	00:09	—	0%	00:01	—	0%
	H36	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
	H37	—	—	0%	—	—	0%	—	—	0%	—	—	0%	16:28	—	10%	—	—	0%
	H38	00:00	—	0%	00:01	—	0%	—	—	0%	00:01	—	0%	00:09	—	0%	00:01	—	0%
	H39	00:00	—	0%	00:01	—	0%	—	—	0%	00:01	—	0%	00:09	—	0%	00:01	—	0%
	H40	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
	H42	00:00	—	0%	00:01	—	0%	—	—	0%	00:01	—	0%	00:09	—	0%	00:01	—	0%
	H43	00:00	—	0%	00:01	—	0%	00:01	—	0%	00:01	—	0%	00:09	—	30%	00:01	—	0%
	H44	00:00	—	0%	00:01	—	0%	00:01	—	0%	00:01	—	0%	00:09	—	0%	00:01	—	0%
	H45	00:06	—	0%	00:31	—	0%	—	—	0%	00:31	—	0%	00:13	—	0%	00:01	—	0%
	H46	—	—	0%	—	—	0%	—	—	0%	—	—	0%	02:08	—	0%	—	—	0%
	H47	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
	H48	—	—	0%	—	—	0%	—	—	0%	—	—	0%	16:27	—	0%	—	—	0%
	H49	—	—	0%	—	—	0%	—	—	0%	—	—	0%	16:28	—	50%	—	—	0%
	H50	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
	H51	00:00	—	0%	00:01	—	0%	00:01	—	0%	00:01	—	0%	00:09	—	0%	00:01	—	0%
zephyr-sam4s	FP_FRB16	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
	FP_FRB17	—	—	10%	06:55	09:21	80%	—	—	30%	04:09	05:48	70%	—	—	40%	00:10	00:12	70%
	FP_FRB43	00:37	—	0%	05:11	—	0%	09:14	—	10%	03:37	—	10%	—	—	20%	00:05	—	20%
	FW49	00:30	00:30	100%	02:11	02:48	100%	03:41	04:21	80%	01:36	02:52	100%	07:09	07:09	90%	00:03	00:03	100%
	FW50	00:28	—	0%	01:51	—	0%	02:53	—	0%	01:35	—	0%	05:11	—	0%	00:03	—	0%
	FW51	00:00	—	0%	00:09	—	0%	00:10	—	0%	00:09	—	0%	—	—	0%	00:01	—	0%
	FW52	—	—	0%	14:05	—	0%	—	—	0%	—	—	0%	—	—	0%	—	—	0%
	H62	—	—	0%	—	—	40%	—	—	30%	—	—	0%	—	—	0%	—	—	0%
zephyr-sampro	FP_FRB18	00:06	01:45	100%	00:06	—	0%	00:06	—	0%	00:06	—	0%	00:05	—	0%	00:01	—	10%
	FP_FRB19	00:01	06:24	90%	00:02	—	0%	00:01	—	0%	00:04	—	0%	00:05	—	0%	00:01	—	0%
	FP_FRB30	00:00	00:00	100%	00:01	00:01	100%	00:01	00:01	100%	00:02	00:02	100%	00:05	00:05	100%	00:01	00:01	100%
	FW46	06:01	—	50%	00:50	15:37	60%	05:02	—	50%	00:51	—	0%	00:07	00:48	100%	00:03	00:39	100%
	FW50	05:56	—	40%	04:34	05:48	100%	06:30	08:13	100%	14:59	—	20%	00:11	00:38	100%	00:08	00:34	100%
	FW51	06:01	—	0%	00:50	—	0%	05:02	—	0%	00:51	—	0%	00:07	—	0%	00:03	—	0%
	FW52	05:57	—	0%	04:34	—	0%	09:21	—	10%	15:00	—	10%	00:11	10:49	80%	00:08	00:18	100%
	H59	—	—	0%	—	—	0%	—	—	0%	—	—	0%	10:35	—	20%	—	—	0%
	H60	00:09	—	0%	00:06	—	0%	00:05	—	0%	00:06	—	0%	00:05	—	0%	00:01	—	0%
	H61	00:03	—	0%	00:03	—	0%	00:02	—	0%	00:05	—	0%	00:05	—	0%	00:01	—	0%

Table 3: Fuzzing Results (Part 3 of 3)